

INSTRUCTIONS FOR USE – ProbeXYZ.dll

User Manual and Technical Guide

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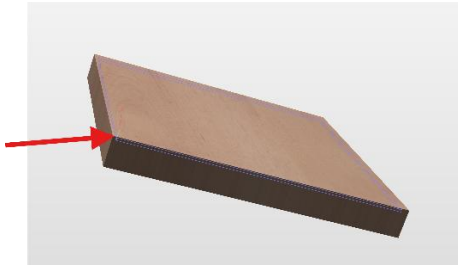
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1. INTRODUCTION

1.1 Purpose and Intent of ProbeXYZ Software

ProbeXYZ is a software developed in conjunction with the popular UCCNC program, which drives several types of CNC machine motion controllers. ProbeXYZ operates exclusively on the Windows operating system within the UCCNC environment as a plugin and cannot function under other CNC software. The purposes of the ProbeXYZ software are:

- To determine the work zero coordinates for the X, Y, and Z axes within the CNC machine's workspace, specifically at the top front-left corner of the material to be machined.



- To determine only the Z-coordinate on the material surface.
- To determine the lower safety software travel limit for the Z-axis (-Z soft limit).

1.2 Technical Specifications and Requirements

ProbeXYZ utilizes a touch plate for probing, which must be connected to a controller input.



A	B	C	D	E
15.00mm	10.00mm	50.00mm	50.00mm	5.00mm
1.5cm/0.59inch	1cm/0.39inch	5cm/1.97inch	5cm/1.97inch	0.5cm/0.2inch



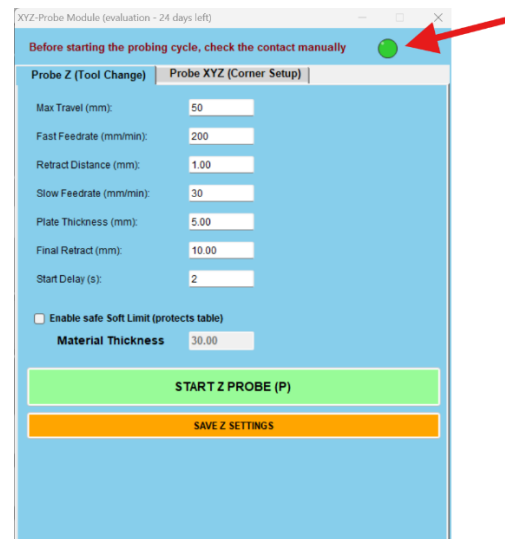
This input must be enabled as Probe 1 or Probe 2 for the plugin to function.



1.3 Important Safety Warnings

Any machine handling carries a potential risk of injury and material damage if basic safety rules and operating instructions are not followed. Before starting the probing process, it is mandatory to familiarize yourself with the following safety measures:

- **Emergency Stop:** Ensure that the E-Stop (the main emergency stop button for the machine) is always within arm's reach.
- **Manual Signal Verification:** Before launching an automatic measurement cycle, make sure to manually connect the probe and the tool to check if the state (signal) changes on the plugin form inside the UCCNC environment, ensuring that the system recognizes the contact.



- **Feed Rate:** During the initial approach to the probe, use a reduced feed rate until you determine the optimal speeds.
 - **Workspace Cleanliness:** The area around the probe and the workpiece being measured must be cleared of wood chips and other debris to ensure that the probe seats perfectly flat on the material.
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2. PREPARATION AND INSTALLATION

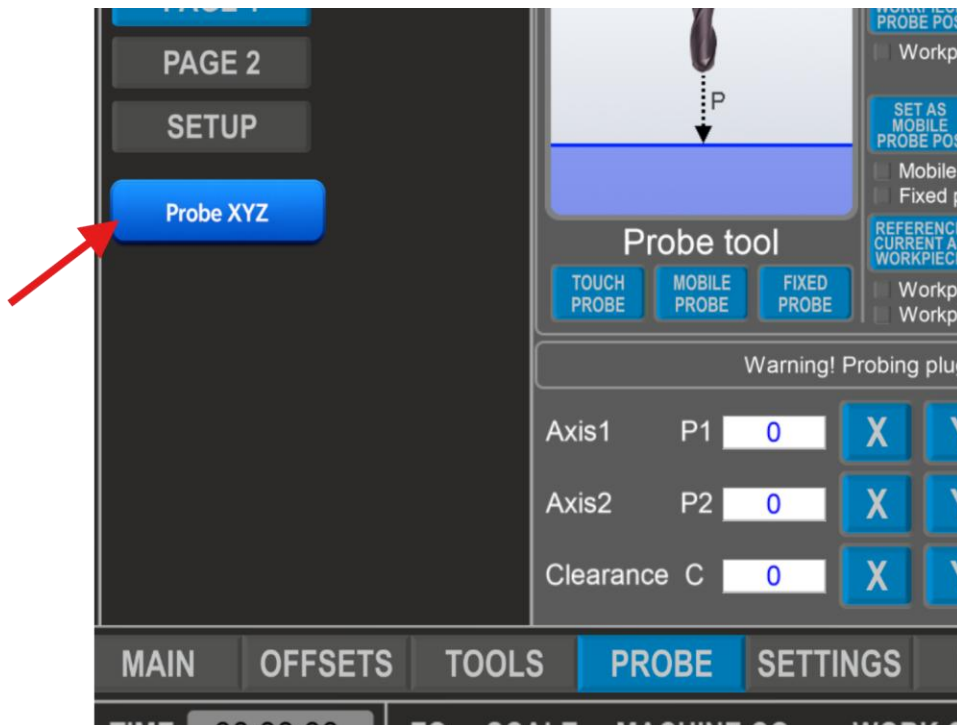
2.1 Package Contents and Installation

The .zip file downloaded from the internet contains the following files:

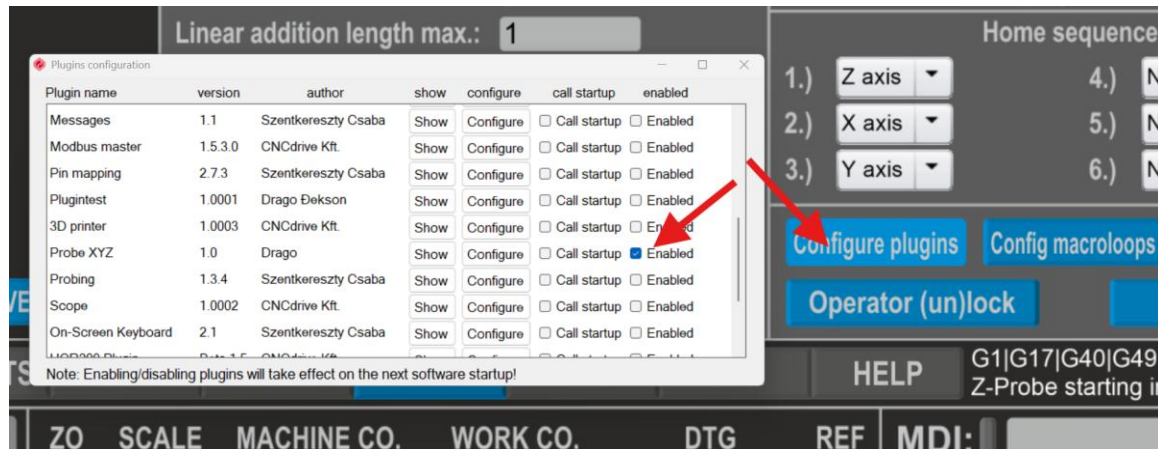
- ProbeXYZ.dll
- Probe_XYZ_instructions_v1_EN.pdf
- ProbeXYZ_button_up.png
- ProbeXYZ_button_down.png

To install, simply copy the **ProbeXYZ.dll** file into the Plugins folder inside the UCCNC directory. The default path is: C:\UCCNC\Plugins.

The .png button image files are provided in case you want to add a shortcut button to the UCCNC screen (for example, inside the PROBE tab).



After copying it to the Plugins directory, the plugin will become visible in the UCCNC plugin list under the name **"Probe XYZ"**. To activate and work with the plugin, it is required to check the **"Enabled"** checkbox and restart the UCCNC software.

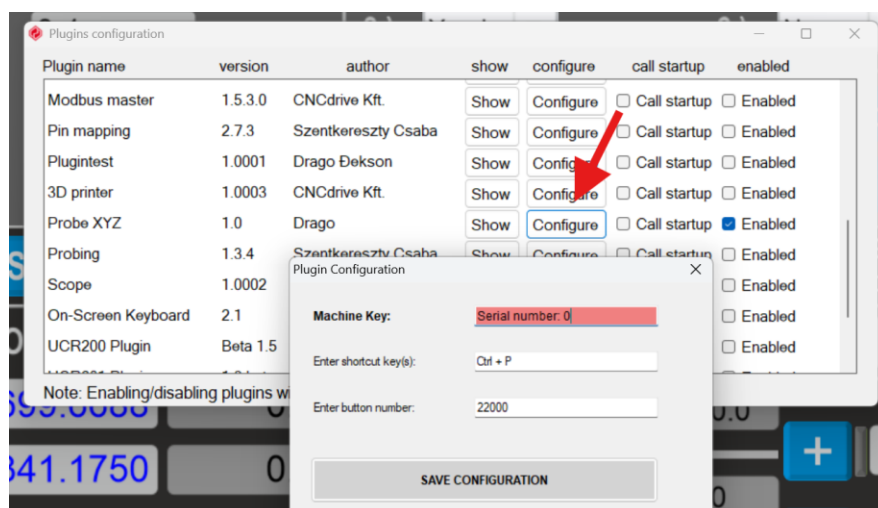


3. FORMS AND CONTROLS

3.1 Configuration Form

The Configuration Form opens by clicking the **"Configure"** button in the UCCNC plugin list form where our plugin is located. The controls allow you to do the following, in order:

- Display the serial number of your controller.
- Enter a hotkey or hotkey combination for quick launching of the plugin.
- Define the **"buttonnumber"** parameter, which is required for creating a shortcut button on the screen to quickly launch the plugin.
- Save the configured parameters into a .pro file.



3.2 Main Form – Probe Z Tab

Every input in the parameter entry fields features data validation; only numbers can be entered, and they must fall within a specific range for each individual parameter. Use a period (dot) as the decimal separator.

XYZ-Probe Module (evaluation - 24 days left)

Before starting the probing cycle, check the contact manually

Probe Z (Tool Change) | Probe XYZ (Corner Setup)

Max Travel (mm): 50

Fast Feedrate (mm/min): 200

Retract Distance (mm): 1.00

Slow Feedrate (mm/min): 30

Plate Thickness (mm): 5.00

Final Retract (mm): 10.00

Start Delay (s): 2

☐ Enable safe Soft Limit (protects table)

Material Thickness 30.00

START Z PROBE (P)

SAVE Z SETTINGS

Parameter	Description	Min value	Max value	Optimal value
Max travel (mm)	Maximum possible tool travel distance until touch-plate contact	1	60	50
Fast Feedrate (mm/min)	Approach speed to the probe before the first touch	5	300	250
Retract Distance (mm)	Retraction distance after the first touch	0.2	3	0.5
Slow Feedrate (mm/min)	Approach speed to the probe before the second touch	5	50	30
Plate Thickness (mm)	Thickness of the touch plate (probe)	0	40	Measure the thickness of your plate
Final Retract (mm)	Z-axis lift height after completion of probing sequences	1	100	5
Start Delay (s)	Delay before probing starts after initiating the sequence	0	15	Long enough for you to reach the touch plate

- **“Enable safe Soft Limit (protects table)”** – explanation at the end of this chapter.
 - **“START Z PROBE (P)” button** – clicking this button or pressing the **“P”** key (if the plugin form has focus) initiates the Z probing sequence.
 - **“SAVE Z SETTINGS” button** – saves the entered parameters into the `.pro` file. Launching the plugin in the future will automatically load the saved parameters from this `.pro` file.
-

3.3 Main Form – Probe XYZ Tab

Every input in the parameter entry fields features data validation; only numbers can be entered, and they must fall within a specific range for each individual parameter. Use a period (dot) as the decimal separator.

The screenshot displays the 'XYZ-Probe Module (evaluation - 21 days left)' window. At the top, a blue header bar contains the text 'Before starting the probing cycle, check the contact manually' and a green status indicator. Below this, there are two tabs: 'Probe Z (Tool Change)' and 'Probe XYZ (Corner Setup)'. The 'Probe XYZ (Corner Setup)' tab is active, showing a list of parameters with input fields: 'Max Travel (mm):' with value 60, 'Fast Feedrate (mm/min):' with value 300, 'Retract Distance (mm):' with value 0.5, 'Slow Feedrate (mm/min):' with value 50, 'Plate Thickness (mm):' with value 5, 'Tool Diameter (mm):' with a dropdown menu showing 6.0000, 'Plate Side Thickness (mm):' with value 10, 'Final Retract (mm):' with value 5, and 'Start Delay (s):' with value 4. Below these fields is a checkbox labeled 'Enable safe Soft Limit (protects table)' which is currently unchecked, and a 'Material Thickness' field with value 40. At the bottom, there are two large buttons: a green 'START FULL XYZ PROBE (P)' button and an orange 'SAVE XYZ SETTINGS' button.

Parameter	Value
Max Travel (mm):	60
Fast Feedrate (mm/min):	300
Retract Distance (mm):	0.5
Slow Feedrate (mm/min):	50
Plate Thickness (mm):	5
Tool Diameter (mm):	6.0000
Plate Side Thickness (mm):	10
Final Retract (mm):	5
Start Delay (s):	4
Material Thickness	40

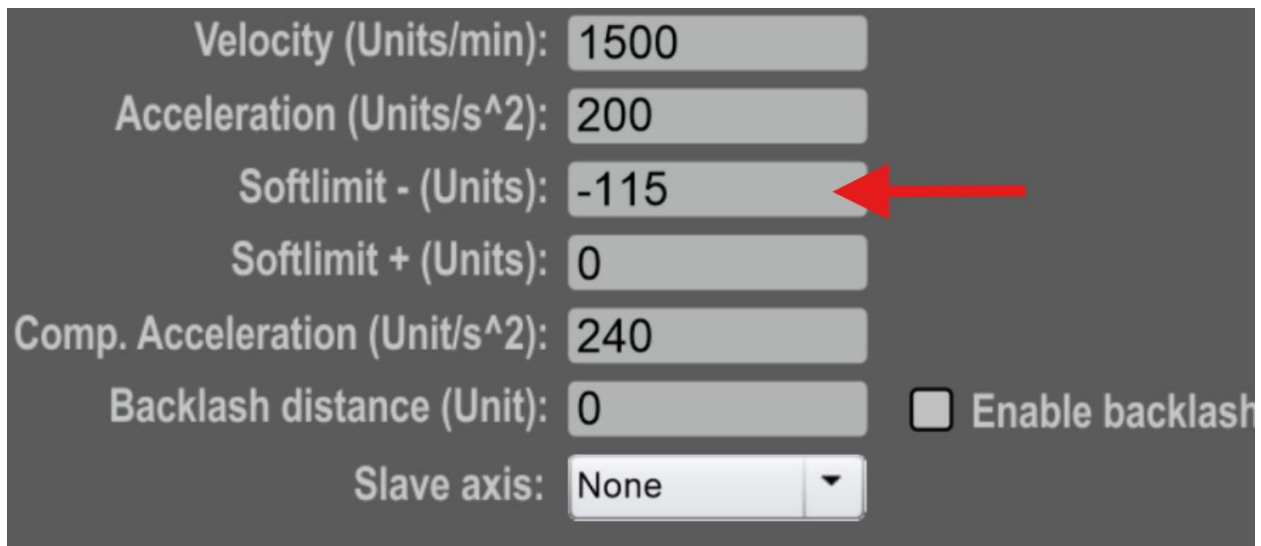
Parameter	Description	Min value	Max value	Optimal value
Max travel (mm)	Maximum possible tool travel distance until touch-plate contact	1	60	50
Fast Feedrate (mm/min)	Approach speed to the probe before the first touch	5	300	250
Retract Distance (mm)	Retraction distance after the first touch	0.2	3	0.5
Slow Feedrate (mm/min)	Approach speed to the probe before the second touch	5	50	30
Plate Thickness (mm)	Thickness of the touch plate (probe)	0	40	Measure the thickness of your plate
Tool Diameter (mm)	Allows entering the tool diameter or selecting a standard inch tool, with automatic conversion to mm	0.1	100	The tool diameter used during probing
Plate Side Thickness (mm)	Side thickness of the touch plate (probe)	0	100	Measure the side thickness of your plate
Final Retract (mm)	Z-axis lift height after completion of probing sequences	1	100	5
Start Delay (s)	Delay before probing starts after initiating the sequence	0	15	Enough time for you to reach the touch plate

- **“Enable safe Soft Limit (protects table)”** – explanation at the end of this chapter.
- **“START FULL XYZ PROBE (P)” button** – clicking this button or pressing the **“P”** key (if the plugin form has focus) initiates the complete XYZ probing cycle.

- **“SAVE XYZ SETTINGS” button** – saves the entered parameters into the .pro file. Launching the plugin in the future will automatically load the saved parameters from this .pro file.
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3.4 Enabling -Z Soft Limits

By checking the **“Enable safe Soft Limit (protects table)”** checkbox, you enable the overwriting of the Z-axis Softlimit value following a Z or XYZ probing sequence. This updates the limit to a new value that completely prevents the tool from plunging deeper than the surface of the machine bed (spoilboard). This new value depends on the thickness of the material currently being machined. The material thickness is entered into the text box next to the **Material Thickness** label, within a range of 0 to 130mm.



The screenshot shows a settings window with the following parameters and values:

Parameter	Value
Velocity (Units/min):	1500
Acceleration (Units/s ²):	200
Softlimit - (Units):	-115
Softlimit + (Units):	0
Comp. Acceleration (Unit/s ²):	240
Backlash distance (Unit):	0
Slave axis:	None

There is an unchecked checkbox labeled "Enable backlash" to the right of the Backlash distance field. A red arrow points to the "Softlimit - (Units)" field.

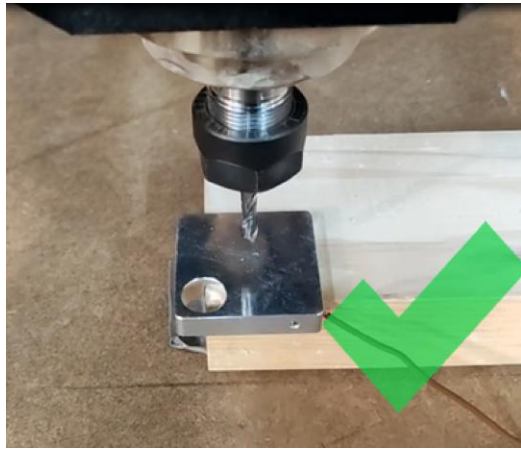
4. PROBING

4.1 General Information

Before probing, it is mandatory to perform a manual connection check (Chapter 1.3). Initiate probing only after confirming that the Probe signal is active and that the plugin registers it via both visual verification (the green LED at the top of the plugin form) and audible confirmation.

By jogging, bring the tool tip approximately 10 to 30mm directly above the center of the touch plate (probe). Do not aim the tool at the hole on the touch plate; the hole is strictly

for visual inspection to confirm that the plate is properly seated on the material, which is true if the material corner is visible in the center of the hole.



The directional descriptions used follow the standard conventions for CNC machines:

- Up and down for the Z-axis,
- Left and right for the X-axis,
- Forward and backward for the Y-axis.

4.2 Z Probing

Steps:

- Set the “**Start Delay**” to the required value; clicking “**START Z PROBE (P)**” starts the countdown (accompanied by a beep) before the machine moves.
- The Z-axis begins moving downward toward the probe at the “**Fast Feedrate**”. The travel distance to the touch plate must be shorter than the “**Max Travel**” parameter.
- Upon contact, the Z-axis retracts (moves away) by the “**Retract Distance**” value.
- The Z-axis descends again at the “**Slow Feedrate**” until it touches the probe. This contact triggers the value entry into “**Z work co**”.
- The Z-axis lifts to the “**Final Retract**” height. Z probing is now complete.

Following a successful probing sequence, the “**Z work co**” value must equal the sum of the absolute values of the “**Plate Thickness**” and the “**Final Retract**”. Always verify this after probing.

4.3 XYZ Probing

Steps:

- Enter the “**Tool Diameter**” value matching the tool you will use for probing. You can also select an imperial/inch standard from the drop-down menu (for automatic inch-to-mm conversion).
- Set the “**Start Delay**” to the required value; clicking “**START FULL XYZ PROBE (P)**” starts the countdown (accompanied by a beep) before the machine moves.
- The Z-axis begins moving downward toward the probe at the “**Fast Feedrate**”. The travel distance to the touch plate must be shorter than the “**Max Travel**” parameter.
- Upon contact, the Z-axis retracts (moves away) by the “**Retract Distance**” value.
- The Z-axis descends again at the “**Slow Feedrate**” until it touches the probe. This contact triggers the value entry into “**Z work co**”.
- The Z-axis lifts to the “**Final Retract**” height. The Z sequence is now complete.
- The X-axis moves left.
- The Z-axis moves down.
- The X-axis moves right at the “**Fast Feedrate**” until it makes contact with the probe.
- Upon contacting the probe, the X-axis briefly retracts to the left by the “**Retract Distance**” value.
- The X-axis moves right at the “**Slow Feedrate**” until it touches the probe. This contact triggers the value entry into “**X work co**”. The X sequence is now complete.
- The X-axis retracts 3 mm to the left, away from the probe.
- The Z-axis lifts by the “**Final Retract**” value.
- The Y-axis moves backward over the front edge of the probe.
- The X-axis moves right into the probe area.
- The Z-axis descends into the probe contact zone.
- The Y-axis moves forward at the “**Fast Feedrate**” until it makes contact with the probe.
- Upon contacting the probe, the Y-axis briefly retracts backward by the “**Retract Distance**” value.
- The Y-axis moves forward at the “**Slow Feedrate**” until it touches the probe. This contact triggers the value entry into “**Y work co**”. The Y sequence is now complete.
- The Y-axis retracts 3 mm backward, away from the probe.
- The Z-axis lifts by the “**Final Retract**” value.

The complete XYZ probing cycle is now finished.

5. SUPPORT AND USER DATA

5.1 Video Instructions and Useful Links

For a step-by-step visual demonstration of the installation, calibration, and probing process, please watch our official video guide on YouTube.

- **Video Tutorial:** <https://www.youtube.com/watch?v=KhVyHSLwYYI>
- **Official Website:** <https://boreagrgic.github.io/BoreaGrgicWeb>

5.2 Technical Support Contact

If you encounter any issues or have questions regarding the ProbeXYZ software, feel free to contact our technical support team via emails:

- **Support Email 1:** drago.grgic3@gmail.com
- **Support Email 2:** drago.grgic2@st.t-com.hr